Microprocessors Laboratory Report

Lab 3: Autonomous Environmental Control System with DHT11 Sensor

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June 2025

1 Introduction

This laboratory exercise implements an autonomous environmental control system using the ARM NUCLEO M4 microcontroller (STM32F411RE). The system demonstrates advanced microcontroller programming concepts including sensor interfacing, dual-mode operation and multi-level alert management.
The system features a DHT11 temperature-humidity sensor interface, touch sensor-driven mode switching, comprehensive UART command interface, and sophisticated alert management with panic reset functionality. The implementation showcases interrupt-driven architecture, real-time environmental monitoring, and user-interactive control systems.
The following sections detail the system architecture, implementation methodology, testing procedures, and insights gained from developing this comprehensive embedded environmental control solution.

2 System Overview

2.1 Hardware Components

The system integrates the following hardware components:

- **DHT11 Sensor:** Connected to GPIO pin `PC_0`, providing digital temperature and humidity measurements using 1-wire communication protocol.
- **Status LED:** Connected to GPIO pin `PB_0`, used for visual alert indication with programmable blinking patterns.
- **Touch Sensor:** Connected to GPIO pin `PC_1`, configured with rising trigger interrupt for mode switching.
- **UART Interface:** Provides bidirectional communication for user authentication, command processing, and data output at 115200 baud rate.

2.2 Software Architecture

2.2.1 Key Data Structures

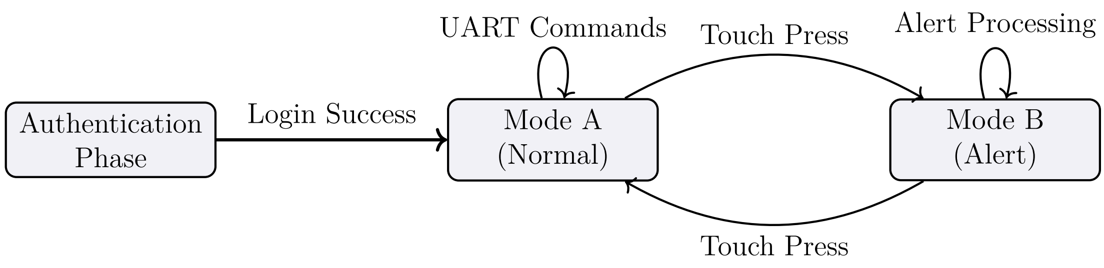
Data Structures

- **System State Management:** Global `volatile` variables manage operational modes, authentication states, and configuration parameters. The `volatile` keyword ensures correct ISR-main loop data sharing.
- **UART Receive Queue:** A `Queue` structure buffers incoming UART characters, decoupling interrupt reception from main loop processing.
- **Sensor Data Storage:** Floating-point variables store the most recent temperature and humidity readings.
- **Authentication Context:** Secure storage for user credentials and AEM (student ID) with proper buffer management and security considerations.

2.2.2 Operational Modes

The system operates using a dual-mode architecture with authentication gate:

- **Authentication Phase**: Password verification followed by AEM capture with secure input handling and visual feedback.
- **Mode A (Normal)**: Standard environmental monitoring with configurable frequency and display options.
- **Mode B (Alert)**: Enhanced monitoring with LED alerts, threshold detection, and automatic recovery mechanisms.



2.2.3 Interrupt Service Routines

Four main interrupt handlers orchestrate system operations:

- **timer_callback()**: Manages periodic sensor readings, LED blinking timing, and system timing base with 1-second resolution.
- **button_press_callback()**: Handles touch sensor events with debouncing, mode switching logic, and frequency update calculations.
- **uart_receive_callback()**: Processes incoming UART characters with queue buffering and ASCII validation.

3 Implementation Details

3.1 UART Command Interface

The system provides comprehensive command processing:

Command	Function	Range/Constraints
'a'	Increase frequency (decrease period)	Minimum: 2 seconds
'b'	Decrease frequency (increase period)	Maximum: 10 seconds
'c'	Toggle display mode	Temperature/Humidity/Both
'd'	Show detailed system status	All parameters
'status'	Compact system information	Quick overview
Ctrl+C	Clear screen and refresh	Terminal reset
Ctrl+X	Safe system termination	Graceful shutdown

3.2 Enhanced Console Interface

The system features a sophisticated UART interface with:

- **ANSI Color Coding:** Different colors for various message types
- **Animated Headers:** Professional-looking bordered headers for messages
- **Progress Indicators:** Visual feedback during authentication and processing
- **Input Restoration:** Smart terminal handling that preserves user input during system messages

4 Testing

Test Scenarios

Test Category	Test Case	Expected Outcome
Authentication	Correct password "2025"	Proceed to AEM input stage
Authentication	Incorrect password	Error message, re-prompt
Authentication	AEM input "10736"	Successful login, menu display
Sensor Reading	Normal conditions	Temperature/humidity display
Mode Switching	Touch sensor press	Toggle between Mode A/B
Frequency Control	Command 'a'	Increase frequency (min 2s)
Frequency Control	Command 'b'	Decrease frequency (max 10s)
AEM Frequency	3rd touch press	Update from AEM last 2 digits
Display Toggle	Command 'c'	Cycle through T/H/Both modes
Alert System	Temp > 25°C in Mode B	LED blinking activation
Alert Recovery	5 normal readings	LED stops, alert cleared
Panic Reset	Temp > 35°C (3 times)	System reset triggered
Status Display	Command 'd' or 'status'	Detailed system information
Terminal Control	Ctrl+C	Screen clear, menu refresh
Safe Shutdown	Ctrl+X	Graceful program termination

4.1 Validation Methodology

Validation Steps

- ✓ **Sensor Accuracy:** Verify DHT11 readings against reference measurements and validate checksum integrity.
- ✓ **Mode Transitions:** Test touch sensor responsiveness and proper state transitions between modes.
- ✓ **Alert Thresholds:** Validate alert activation/deactivation with controlled environmental conditions.
- ✓ **Command Processing:** Verify all UART commands function correctly with proper error handling.

4.2 Challenges and Solutions

Implementation Challenges

1. **DHT11 Timing Critical Sections:** The DHT11 sensor requires microsecond-precise timing for proper communication. Any interrupt during the critical reading phase could disrupt the timing and cause data corruption or communication failure.

   Solution: Implemented critical sections during DHT11 communication by temporarily disabling interrupts using `__disable_irq()` and `__enable_irq()` around timing-sensitive operations. Additionally, used dedicated timing functions with precise delay mechanisms to ensure protocol compliance. The sensor reading is scheduled during low-activity periods to minimize interrupt conflicts.

5 Conclusion

This laboratory project successfully demonstrates the implementation of a sophisticated autonomous environmental control system. The solution integrates multiple advanced concepts including sensor interfacing, dual-mode operation, secure authentication, and intelligent alert management.

Key achievements include robust DHT11 sensor communication with precise timing control, comprehensive user authentication with security considerations, flexible operational modes with touch sensor control, and a multi-level alert system with automatic recovery capabilities.

The implementation showcases professional embedded software development practices including modular architecture, interrupt-driven design, proper error handling, and user-friendly interface design. The system provides a solid foundation for real-world environmental monitoring applications with potential extensions for IoT connectivity and advanced analytics.

References

- ▷ STM32F411RE Microcontroller Reference Manual
- ▷ DHT11 Temperature-Humidity Sensor Datasheet
- ▷ Touch Sensor Datasheet